



**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450/ITS432
EXAMINATION	:	JULY 2023
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
 - i) the Question Paper
 - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

QUESTION 1

- a) Explain **TWO (2)** limitations of the file-based approach in business environment.
(4 marks)
- b) Data models are used as a communication tool between database designers, programmers and users. Identify and briefly explain any **TWO (2)** basic building blocks used in data models.
(4 marks)
- c) Table 1 shows the sample data for an art and culture club in Jasin city.

StudID	Name	Name	Skill
2023123456	Izyan Mimiey	Ahmad	Modern Dance, Singing
2023456789	Jeonghan	Lee	Traditional Dance
2023455abc	Hafizal	Azri	Modern Dance, Playing Drum

- i) Explain **THREE (3)** characteristics of a relational table that have been violated in Table 1.
(6 marks)
- ii) As a database designer, briefly explain possible courses of action to handle each problem stated in (i).
(6 marks)

QUESTION 2

- a) Database Life Cycle (DBLC) is a properly designed framework for implementing a database system. The proper database design leads to an efficient data retrieval. Briefly explain **FOUR (4)** activities in the database design phase.
(8 marks)
- b) XYZ Technologies Sdn. Bhd. is planning to implement a mobile database for their company due to increasing branches in Malaysia and Indonesia. Do you think that mobile database is suitable for this company? Briefly describe **THERE (3)** factors that contribute to the wide usage of mobile database.
(8 marks)
- c) Data is a raw and unorganized fact that required to be processed to make it meaningful. State the differences between structured and unstructured data.
(4 marks)

QUESTION 3

Figure 1 shows an example of banquet reservation form for Dewan Cempaka.

Banquet Reservation Form		
Dewan Cempaka		Reservation ID : 12345 Reservation Date : _____
Details:		
Name :		
Email :		
Address :		
Phone Number :		
Date of Banquet		
Type of Event :		
Number of Guests :		
Arrival Time :		
Food Service Time :		
Banquet Menu and Facilities:		
Menu Code and Name	Menu Description	Quantity
1.		
2.		
Facility Code and Name (You can only choose one facility)	Facility Description	
1.		
Total Payment : _____		
Payment Type (Choose one): ☐ Cash ☐ Credit Card ☐ Check		
THANK YOU!		

Figure 1 : Banquet Reservation Form

Based on the given reservation form, answer the following questions:

- a) Define normalization and explain the purpose of having normalization in a database table.
(3 marks)
- b) Describe **THREE (3)** effects of a poor database design.
(6 marks)
- c) Show all the steps to normalize the structure until third normal form (3NF). The primary key(s), partial dependencies and/or transitive dependencies need to be clearly identified.
(11 marks)

QUESTION 4

The relation schema below shows the loan database system.

DEPARTMENT	(<u>DeptID</u> , DeptName)
APPLYLOAN	(<u>ClientNum</u> , <u>LoanID</u> , LoanTotal, LoanRemainder, LoanDate)
LOAN	(<u>LoanID</u> , LoanType, StaffID)
CLIENT	(<u>ClientNum</u> , ClientName, ClientAddress, ClientPhoneNum)
OFFICER	(<u>StaffID</u> , StaffName, DeptID)

Write SQL statements based on the following requirements.

- a) Create the APPLYLOAN table.
(4 marks)
- b) By using subquery, produce a list of client's names that makes a personal loan.
(4 marks)
- c) Generate the total loan from year 2021 until 2023.
(4 marks)
- d) List the names of officer who manage the housing loan and they are from finance department.
(4 marks)
- e) Generate the total loan for each loan type.
(4 marks)

QUESTION 5

Bristledbeast is a hiking agency that manages various trips to hike local mountains. They plan to build a database system in order to keep track the organized trips throughout the years. You are hired to design and develop a database system based on the requirements needed are as following:

- Details of hikers need to be recorded in the system. Hiker identification number, full name, address, phone number, email.
- Each hiker also needs to give details of any family member. They need to give their family member's full name, emergency contact number, address and email.
- For each trip, the trip ID, trip name, approximate length (in hours), and fees charged are recorded in this table.
- Each hiker can join many trips. Start date of hike and end date of hike are also need to be recorded in this system.
- Each trip is managed by at least one guide. Some guides have been assigned to several trips or may not assign to any trip. Payment for each assigned trip is recorded in this system. Guides are identified by an employee ID. Guide full name, contact number, address, hire date are also recorded in this system.
- Each trip will have one destination of mountain. The details of mountain such as mountain name, height and country are also need to be stored in this system.

Based on the given information:

- a) Construct an Entity Relationship Diagram (ERD) complete with all the required tables, attributes, relationships as well as the cardinalities using the Crow's Foot notation that reflects all of the given business rules.

(16 marks)

- b) Briefly explain **TWO (2)** examples of reports that can be generated from ERD in (a).

(4 marks)

END OF QUESTION PAPER